

Progress Report: Prompt Design Guideline Formation by Experimentation on Prompt Structures for AI Art

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ABSTRACT

Prompt engineering is the process of creating a suitable phrase to input in a model and process that information to output some sort of media. In the case of AI text to image generators like Stable Diffusion and DALL-E-2 process these prompts to form images. Due to recent evolution of these models, there are not enough guidelines available for users like artists who find it time consuming to form a correct prompt to output their desired image. Thus, research is made on several prompt structures to test out which one performs best with different subjects, styles and mediums by conducting a survey of generated images to digital art students to gather their feedback. Results will be evaluated to reach a conclusion and provide guidelines on prompt engineering for AI art generators.

1. INTRODUCTION

1.1 Problem Definition

A prompt is a textual phrase that identifies what the language model should output, e.g., “a dark forest” would be given to a text-to-image generator to output that image [1, 2]. The public, more specifically artists and designers should have a solid foundation and understanding of effective prompt design for AI art text to image generators. Unfortunately, most of the companies who created these models do not provide an in-depth guideline on prompt engineering for their generator since the relationship between the text and the image being generated is still yet to be completely understood. Not even the developers know entirely the best prompt structure to use as data is scraped from multiple internet sources [3, 4], thus, experimenting with the model and using the generator is the best way to test and deduce a form of manual or guidelines.

Due to recent developments of AI art generators not many people know how to write their prompt yet, which leads to a method of trial and error [5, 6]. This is very time consuming since the user would adjust the prompt repeatedly until their desired image is generated and thus sheds light on requiring guidelines on prompt engineering. This problem is appearing in the current work scene of artists like Mr Schiavone, an Italian VFX compositor who works in high-end feature films like Marvel and DC movies where in a held interview stated that no prompt guidelines are provided by the movie production and instead let the artists wonder on their own by searching tools, blogs, articles and videos on good prompts that lead to an aesthetically pleasant image.

1.2 Motivation

The dispersed production of image generators like DALL-E-2, Stable Diffusion, Midjourney and Disco Diffusion [7, 8, 9, 10] have led to scarce research on prompt engineering to find the best structure and keywords in the prompt that leads to the best-wished outcome. This area is gaining much popularity recently and several communities are being formed that share images with their prompts to inspire and learn from other designers [11]. Artists like Mr Schiavone are very impressed with the potential of these AI art generators and explain how these are already being used in movie production for little secondary tasks that are time consuming due to control and stability. Thus, this FYP would help further expose how prompting works and increase knowledge for these people. Mr Schiavone continues to add how these AI generators are becoming tools to boost production time and are setting a standard to express creativity just like when ‘Photoshop’ was introduced. This FYP would be part of a new evolution in the creative AI scene that would inspire and help others ease in this process of prompt engineering.

2. AIMS AND OBJECTIVES

This FYP will provide an in depth analysis on a current area of research in the AI creative space by performing an experiment and compare results with other research papers and current AI models. No final deliverable or program will be resulted from this project.

Thus, this FYP aims to shed light on a rather new and evolving topic by analysing which prompt structure produces the best resultant image as it is important for users and developers to know and so acts as a guideline for prompt engineering.

The main objectives are:

- Investigating and gathering information on the foundation of prompts and how they are being used in current AI image generators like Stable Diffusion and DALL-E-2 by research, involvement in communities and interviews
- Using these models and experimenting with various forms of prompts to generate specific images
- Validating and evaluating samples of the images generated with their prompts together with Digital Art students by forming a survey with a self made Web application.
- Feedback given will be used to reach a conclusion and provide statistics to finally create the guideline and result in the most reliable prompt structure.

3. BACKGROUND

3.1 Prompt Engineering

Prompt engineering is a way of forming and structuring phrases or sentences to achieve some form of styled synthesis. This does not necessarily have to do with text-to-image generation only as it is applied to multiple other applications. The concept of prompt engineering was standardised by the GPT-3 project, Open AI's language model where prompts could be used to generate essays, code, write emails and more [12]. Open AI helps users with prompt design by following their brief guidelines which advise that portraying what you want to output and not just telling the generator the image you want, is the secret to a good prompt [13]. This means that specifically inputting any necessary detail is better than providing a very brief phrase as the more information the better the output is.

3.2 AI Art Communities

As prompt design and image generation are gaining popularity several communities are being formed online, especially on platforms like Twitter, Reddit and Discord [14, 15, 16, 17, 18, 19, 20]. People are sharing their creations to inspire others and discover any new keywords which might affect the output from the model. Not all content shared is annotated by its prompt so the research made will delve into what kind of prompts are found online when interacting with other members of the community. A text-to-image generator, Midjourney is solely based on Discord [18] where the prompt can be entered in a group chat and the image is generated there where multiple people can see it. This allows for the greatest form of knowledge sharing since the prompts can be seen by everyone.

3.3 AI Features

3.3.1 CLIP

CLIP is a multimodal model developed by OpenAI that is trained on 400 million captioned images scraped throughout the internet [3]. Its function is to understand how strong the relationship between the prompt and the image is, that is, CLIP would provide the strongest prompt when inputting a certain image that best matches the prompt. This contradicts other models where the focus is predicting labels for images by the use of classifiers. CLIP avoids classifiers by using zero-shot learning which is being able to classify a class that was never seen before in the training set. This allows for vast possibilities and is an advantage to others since no specific classifiers need to be trained

3.3.2 Diffusion

The process of generating images is called diffusion. Diffusion, inspired by non-equilibrium thermodynamics is the process of introducing gaussian noise to an image at each time step until the image becomes filled with Gaussian noise. The main part of training is to reverse this diffusion process by de-noising the image and creating a new form of the image at each time step until the image is finally generated with partly or completely new additions according to the prompt provided. The structure of diffusion is guided by a Markov chain to allow for a sequential process of noising and de-noising the image [21, 22, 23]. The AI model would then be trained to perform this de-noising process to a noised image to be able to generate an image influenced by the prompt.

A variation of the diffusion model is the latent diffusion model. This is the one used by Stable Diffusion where the de-noising stage of the diffusion process is accomplished in a latent space [23]. The aim of this model is to achieve the same result by working in a compressed image state which would be the latent space. This would avoid processing images in the pixel space which is rather large. Thus, downsampling the task would make the process much more efficient and with the combination of encoders and decoders would help to take the most relevant information of the image whilst reducing the size where at the end the image is upsampled back again to view the image as intended [24].

3.3.3 GANs for Generative AI

Before diffusion models rose to popularity, generative adversarial networks were used to generate mostly images and videos. GANs are computational structures that employ two neural networks, pitting one against the other to produce new, synthetic instances of data that can pass for real data. The first neural network is called the generator which generates new data and the second neural network is called the discriminator which evaluates that generated data on whether it should be part of the training set [25]. This process would then be repeated until the model creates an understanding to distinguish elements between each other. This type of model is being used less nowadays due to being unstable as it is prone to overfitting which results in generating the same image for several prompts and other models like diffusion are generating more realistic images with an evened distribution. [26].

3.3.4 Training Models

The models that make use of prompting require a prompt dataset for training in order to accomplish a task. In the case of text-to image synthesis multiple captions are assigned to an image where the training phase would allow the model to associate the relationship between like CLIP [3]. Datasets like LAION-Aesthetics and COCO [4, 27] contain image captions which are used to train AI art generators. The models formed heavily depend on the prompt given since a simple change in the phrase would result in a different relationship and understanding of the text prompt with the resultant image. Thus this FYP would help the prompt tuning iterative process in the training phase since which prompt structures works best and how each one differs would be analysed.

4. LITERATURE REVIEW

4.1 Current Prompt Guidelines

The official websites of these text-to-image generators provide their own variation on prompt design tips or guidelines.

4.1.1 Midjourney

The Midjourney guide advises users that creating ambiguous prompts is a good approach to acquire diversity, but it might not provide what the user wants. That said, they encourage users to be as general as they like since anything left out from the prompt will be randomised to promote versatility in aspects of the image. On the other hand, a prompt can be detailed by pointing out attributes of an image like "three monkeys in business suits" instead of "monkeys doing business" as Midjourney advises to be as specific as possible about any

background or information that is relevant to the image to be generated [28].

The guide continues to add several tips on prompts such as, describing a style e.g., “a surreal landscape”, mentioning artists to provide a specific style, avoiding concepts that involve significant extrapolation e.g., “an optimistic vision of an augmented reality future” and using singular nouns or specific numbers [28]. The tips provided by Midjourney are a useful addition to the research made in this FYP. Taking advantage of knowing what the model understands together with its limitations is a plus for prompt engineering. Thus, with the focus of prompt structures being analysed in this FYP would certainly aid this field of tips for prompt design in the AI creative space.

4.1.2 DALLE-2

This guide starts by defining a prompt to be a statement of 400 characters or fewer that explains the image you desire [29]. No guide mentioned in this FYP has purposely stated the prompt character limit except for the DALLE-2 prompt book. This could be the case that the developers rigorously tested the model and found out that it would not pick up the remainder of the words for the image generation process. In this FYP prompts of fewer characters will be tested due to feasibility issues as long prompts are rather inefficient to analyse and evaluate.

The guide emphasises that the model reaches its own decisions as it is trained on millions of images with their captions. Thus, not even the developers themselves can create a concrete manual for prompt design since this can only be derived by using the model and discovering how it understands the relationship of text with images [29]. This FYP will offer insight on whether distinct prompt structures provide different image results being better or worse by using each model while analysing and comparing the output. This form of research and experimentation will be a step closer to the concrete manual designed for prompt engineering.

4.1.3 Stable Diffusion

DreamStudio is an external tool developed by the same creators of Stable Diffusion [30] that uses their model to provide a graphical user interface for generating images. DreamStudio provides a simple documentation for prompt creation when compared to others. The guide explains how raw prompts are prone to randomly generated images since the prompt consists only of a few words e.g., “panda” and “a warrior with a sword”. This is the foundation of any prompt where it involves the subject of the sentence, that is the focus of the image [31]. This FYP will place focus on selected thought out subjects to portray a variety of generations where the foundation of a prompt structure will be experimented upon.

The guide continues to clarify how style is a critical criterion of a prompt since it effects the image the most after the raw prompt. If no style is mentioned in the prompt the model would take the style that it has seen the most in similar images. Examples of how a certain style can be implemented in a prompt are “a photo of + raw prompt”, and “a photograph of + raw prompt” [31]. A variety of styles will be thoughtfully picked out when performing the experiment of this FYP to

assess how each style is represented with different subjects and how would the resultant image differ when using a different prompt structure for the same style.

Stability AI provide their own prompt book where they indicate that a prompt is limited to 500 characters [30, 32] which is 100 less of what OpenAI offers for DALLE-2 as mentioned previously. The limit of number of characters is not important for this FYP since less characters will be used. The prompt book mentions weighting where certain keywords of a prompt can be boosted or demoted by adjusting weights. This is usually between -2 and 2 for example “beautiful:1.5” and “ugly:-1.0”. The process of adjusting weights to change features of an image is called Multi-Prompting [32]. Multi-Prompting was considered to be included in this FYP, but it was then disregarded since it would be considered as another research proposal.

4.2 Prompt Anatomy

Openlaender studied what a prompt consists of by splitting it into several categories [33]. The author first performs an ethnographic study of the text based generative art community on Twitter. This means that the author took part in discussions involved with the community and witnessed several examples of prompts. This is done to familiarise with how people create their prompts together with their result posted on social media. A similar path was taken for this FYP where several posts of popular Twitter accounts that use these generators were followed and reviewed whilst taking part in discussions of Discord servers for DALLE-2, Midjourney and Stable Diffusion. The author continues to express how text-to-image synthesis is not a new area of research, but text-based generative art is a very recent phenomenon. Thus, there is not much scholarly literature practicing this area yet and so most of the research is based on grey literature like articles on the web. This is statement is still valid as no new papers regarding this area has been published yet apart from [33].

- Subjects – fundamental element of a prompt, the focus of the image.
- Style modifiers – short phrases to promote a style to the image e.g., artist names
- Image prompts – images can be inserted to the model instead of text to enhance or alter a new form of the image.
- Prompt modifiers/Quality boosters – keywords that are added to the prompt to improve and make the image more appealing to the user.
- Repetition – repetition of certain words can be introduced to a prompt to emphasise what the user wants in the generated image.
- Magic terms – those phrases that depict an unreal or imaginary image.

This FYP will make use of subjects, medium and style. The others, more specifically the prompt modifiers were disregarded from the experiment since they are considered as additions, that is being an extra part of the prompt which is optional. Testing is based on the foundation of the prompt structure

and thus the root of the problem would be tackled. In fact, the results in the paper show that the iterative process of creating a good prompt mainly depends on the subject and the style. The author uses ‘VQGAN+CLIP’ as a model to produce images. In this case DALLE-2 and Stable Diffusion will be used as they provide a better output and are more of recent production rather than VQGAN+CLIP which is one of the first text-to-image synthesis generators out there for generative art [34].

Finally, the author includes a future research section. Two of the proposed research questions will be tackled in this FYP. The first question is, “What are the factors that make an input prompt successful in generating a ‘beautiful’ AI-generated artwork?” [33]. This question will be answered with respect to which prompt structure provides the best output. The second question is, “What are the prompt engineering workflows and strategies adopted by practitioners in the text-based generative art community?”. This question will be answered by interviewing current artists working in different scenes like movie production and composers.

4.3 Prompt Design

Liu and Chilton studied and experimented on 5 aspects of prompt engineering that lead to a set of design guidelines [35]. The generator used for their experiments was ‘VQGAN+CLIP’ which is the same one used in [33]. This paper was published in 2021 and [33] was published in early 2022 so the recent evolution of generators like DALLE-2 and Stable Diffusion were not yet made public. Thus, this FYP will take advantage of these new models to be able to compare them with each other and provide guidelines for current users who use these generators.

The first experiment is crucial since it relates most to the goal of this FYP. The authors gathered 9 prompt structures including style, subject and medium where a total of 1296 images were generated. A list of words relating to style, subject and medium were first selected to then form the permutations of each word with the prompt structure. The authors do not specify why and how they chose their selected subjects but only explained that the balance of words being abstract, and concrete was considered as assignment was based on a dataset of concreteness values. Styles were selected based on being figurative and abstract [35]. The process of how mediums were selected is not mentioned in this experiment. In this FYP a similar structured experiment will take place where 7 prompt structures will be used together with a list of subjects, medium and style. The process of how each word was selected is mentioned in the Methodology section.

The goal from these images was to let the selected annotators decide which images looks better or worse and which image stands out from the rest between the prompt structures. The number of annotators called up was only 2 [35]. This is a poor selection and is not enough to support the experiment and thus in this FYP annotators will be students from the department of Digital Arts, reaching a number close to 16. The authors discovered no meaningful difference between different prompt permutations which may be due to very few annotators. This FYP will check if the same result arises for new generators like

DALLE-2 and Stable Diffusion. Thus, a comparison could be then made with Liu and Chilton’s research.

5. PROPOSED SOLUTION

To find out which prompt structure suits a good output image, and which one stands out from the rest, 6 of them will be used for experimentation. These prompts were gathered and formed by viewing several articles [1, 2], community posts and being inspired by the experiment of Liu and Chilton [35]. These are:

[Medium] of [Subject] in the style of [Style]

[Style] [Medium] of [Subject]

[Medium] of [Subject], [Style]

[Medium] of [Subject] with [Style] style

[Medium], [Subject], [Style]

[Subject], [Medium], [Style]

Each prompt includes a subject, style and medium. Each category contains 3 words relating to it where a total of 162 images of different combination of keywords for each model will be generated. This process will be done two times, one for Stable Diffusion and the other for DALLE-2. Thus, 324 images will be generated.

5.1 Process of producing Prompt Categories

Subjects were chosen by first using a dataset of Stable Diffusion and Midjourney prompts acquired from user’s prompt inputs [36, 37] and passing them through an 3 algorithms created by the author of this FYP. The main algorithm, Find Subject algorithm, filters the prompts to only show ones with “of” since the word after that phrase results in the subject.

5.1.1 Filter Count Algorithm - Pseudocode

Filtering the dataset of prompts by choosing only prompts with 3 or more words.

```
prompt_list ← list_of_prompts
list_length ← len(prompt set)
filterCount(prompt_list, new_list):
    for(i ← 0 to list_length):
        if list_length ≥ 3:
            new_list.append(prompt_list[i])
```

5.1.2 Filter Stop Words Algorithm - Pseudocode

Filtering the dataset of prompts by removing unnecessary words like stop words.

```
filterStopWords(prompts):
    stop words ← list of english stop words
    return word in prompts if not in stop words
```

5.1.3 Find Subject Algorithm - Pseudocode

Filtering the dataset of prompts by choosing phrases that include the word 'of'.

```
of_prompt_list ← empty_list
filterOf(prompts, of_list):
    for(i ← 0 to len(prompts)):
        for(j ← 0 to len(prompts)):
            if prompts[j] == "of":
                found ← true
                index ← j
            if(found == true):
                if( (prompt_count - 1) == index ):
                    of_prompt_list.append(prompt(index))
                if( (prompt_count - 1) - index == 1):
                    of_prompt_list.append( prompt[index] +
prompt[index+1] )
                if( (prompt_count - 1) - index == 2 ):
                    of_prompt_list.append(prompt[index] +
prompt[index+1] + prompt[index+2])
            else:
                continue
```

5.1.4 Results

The algorithm outputs the 7 most popular subjects. The results are:

Stable Diffusion: *girl: 100, woman: 85, forest: 56, legends: 49, princess: 46, zeus: 46, city: 42*

Midjourney: *wire: 128, European: 120, intertwined: 97, porcelain: 95, town: 87, thin:82, madness:78*

5.1.5 Selection of Prompt Categories

From these results 3 subjects were formed, namely, *person, forest, dog*. Although not all these subjects are not in the resulted list, they were inspired by them, and a universal approach was chosen to select the subjects.

The styles chosen are based on the book [38]. These styles were selected thoughtfully by choosing art movements of different ages of art periods. These are: *Baroque, Neo-Plasticism, digital art*.

The mediums were picked by analysing several prompts from users in the community [29, 32] and choosing the most popular ones. These are: *painting, photograph, 3d render*.

5.2 Survey Creation

Each image generated will be formed in a survey which is then given to a group of annotators that will be a class of Digital Art students. The survey will be formed by creating a web application which is able to provide a unique URL to each student that gives a unique set of images to each student. This is done to fasten the survey process and split up annotation amongst the students. These will mark which image acts as an outlier and which one looks best aesthetically for each combination of subject, style and medium for each prompt

structure. They will also mark which image best represents the style and medium.

6. EVALUATION PLAN

The goal is to receive the data from the survey provided by the art students and perform analysis on which prompt is more reliable than the other and find out the best prompt structure amongst the 6. A comparison will then be made by the results obtained from images of Stable Diffusion and DALLÉ-2 to check if the experiment results in the same prompt structure being selected as the best for both generators. If not, the difference of why different prompts was chosen will be analysed. Graphs and diagrams of the data received will be presented together with further metrics to justify the analysis and evaluation.

7. CONCLUSION

This report presents research and planning done to accomplish the FYP. The report has outlined the problem risen in the creative and AI scene whilst providing a solid understanding of the current research and information available throughout the web. This is done through research, practical use of models, and interviewing an artist. A well thought out plan for the proposed solution was provided which includes already created data to prepare for the survey formed by the web application yet to be created and distributed to the Digital Art students. The data received would then be evaluated appropriately by supporting the analysis by diagrams, graphs and using metrics.

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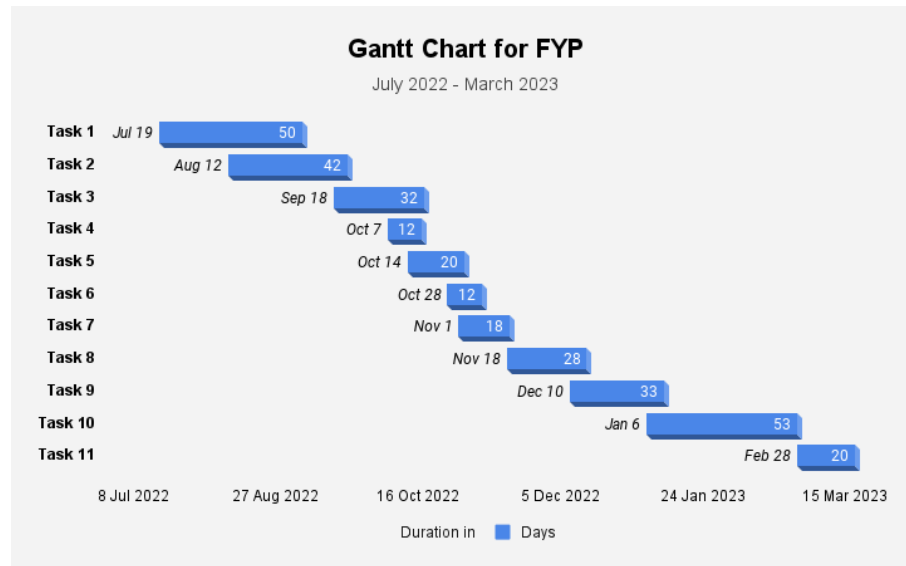


Figure 1: Gantt Chart for FYP Process

Key	
Task	Description
1	Researching various articles and papers to understand the area
2	Test out models by using them (e.g. DALLE-2 and Stable Diffusion)
3	Take part in communities and ask questions like (Stable Diffusion Discord)
4	Find a niche and choose the specific area to work on
5	Start Writing First 3 Chapters
6	Perform Interviews
7	Form methodology Plan
8	Generate images for survey
9	Create web app for survey
10	Write rest of the chapters (Methodology, Evaluation, Conclusion) once feedback from survey is received
11	Finalising FYP write up

Figure 2: Key for Gantt Chart